MCQ A · consolidation (after S7)

After S7: first induction proofs.

For each question, only one option is correct. Justify your choice orally before looking at the solutions.

1. If $u_0 = 2$ and $u_{n+1} = u_n + 3$, then u_2 is:

A. 5 B. 8

C. 11 D. 2^4

2. To prove $P(n)$ by induction from 0, the inductive step consists of:

A. assume $P(n+1)$ B. only check $P(0)$

C. assume $P(n)$ then prove $P(n+1)$ D. solve $P(n) = 0$

3. The sum $\sum_{k=0}^{4} 2^k$ includes:

A. 4 terms B. 5 terms

C. 8 terms D. 2^4 terms

4. If $u_{n+1} - u_n > 0$ for every n, then:

A. u is bounded above B. u is increasing

C. u converges D. $u_n > 0$

5. In `while u <= 10: u = u + 1`, the loop stops when:

A. $u = 10$ B. $u < 10$

C. $u > 10$ D. $u < 0$

MCQ B · review and choice of method (after S20)

After S20: monotone convergence and auxiliary sequence.

For each question, only one option is correct. Justify your choice orally before looking at the solutions.

6. A sequence is increasing and bounded above by 7. We can conclude that it:

A. tends to 7 B. converges

C. is positive D. is geometric

7. For $u_{n+1} = 0{,}5u_n + 4$, any equilibrium value satisfies:

A. $L = 0{,}5L + 4$ B. $L = u_0$

C. $L = 0{,}5 + 4L$ D. $L = 4^n$

8. If $v_n = u_n - 8$ and $u_{n+1} = 0{,}5u_n + 4$, then:

A. $v_{n+1} = v_n + 4$ B. $v_{n+1} = 0{,}5v_n$

C. $v_{n+1} = 8v_n$ D. $v_{n+1} = u_n$

9. "u_n belongs to $]5{,}99; 6{,}01[$" is equivalent to:

A. $u_n < 6$ B. $|u_n - 6| < 0{,}01$

C. $|u_n| < 6$ D. $u_n > 0{,}01$

10. A simulation of 20 terms proves that:

A. the statement is true for every n B. the sequence converges

C. the 20 calculated terms have the observed property D. induction is unnecessary

True or false · justification expected. For each statement, indicate whether it is true or false, then write a proof or a counterexample.

1. Every increasing sequence converges.

2. If $u_n = 6 - 2^{-n}$, then $u_n < 6$ for all n.

3. Solve $L = f(L)$ is enough to prove that every sequence $u_{n+1} = f(u_n)$ converges.

Review of prior work

I can compute a term, a sum, set up an induction proof and study monotonicity using $u_{n+1} - u_n$.
To watch out for: confusing an index with a value; turning a numerical observation into a proof.